PROPERNESS FROM THE DVR VALUATIVE CRITERION OVER A NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4918-4984, following the statement at 4896-4916.

Editorial note. The single '(details omitted)' covers replacing X' by the disjoint union of the irreducible components dominating those of X, flagged in the notes as standard bookkeeping.

Editorial note. The footnote on the sharper formulation (existence after an extension of discrete valuation rings) is transcribed; it does not change the equivalence.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 4962 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Characterization of proper morphisms of spaces. Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-proper (cross-chapter, cited at ~4919-4920). Equivalences (1) iff (2) iff (3).
- Separation from the dvr valuative criterion. lemma-Noetherian-dvr-valuative-separation (same chapter, cited at ~4925). f is separated; reduces to universally closed.
- Universally closed is etale-local on the base. Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-local (cross-chapter, cited at ~4928-4931). Reduces to the affine etale base change V x_Y X -> V.
- Weak Chow's lemma for algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-weak-chow (cross-chapter, cited at ~4958-4960). Quasi-projective X' over Y with a proper surjective X' -> X.
- Image of a proper morphism is proper. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-proper-is-proper (cross-chapter, cited at ~4966-4967). X is proper over Y once X' -> Y is proper.
- Scheme-level dvr valuative criterion. Limits, Lemma limits-lemma-Noetherian-dvr-valuative-proper (cross-chapter, cited at ~4968-4969). Criterion for X' -> Y proper in terms of generic residue fields.

Source reference. Original equation 0H1V: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4818–4818. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0H1W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4829–4829. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H1X: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089J: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IX: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A40: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08AJ: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IT: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06NH: focused-seeds/geometry/S9-spaces-limits-Noetherian-dvr-valuative-proper/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-separated and of finite type and Y is locally Noetherian. The following are equivalent:*

(1) *f is proper,*

(2) *f satisfies the valuative criterion, see Morphisms of Spaces, Definition 03IX,*
(3) *for any diagram (0H1V) there exists exactly one dotted arrow,*
(4) *for all diagrams (0H1V) with A a discrete valuation ring there exists exactly one dotted arrow, and*
(5) *for all diagrams (0H1V) where A is a discrete valuation ring and where the image of* $\mathrm{Spec}(K) \to X$ *is a point of codimension* 0 *on X there exists exactly one dotted arrow*[1].

Proof. We have (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) by Morphisms of Spaces, Lemma 0A40. It is clear that (3) $\Rightarrow$ (4) $\Rightarrow$ (5). To finish the proof we will now show (5) implies (1).

Assume (5). By Lemma 0H1W we see that f is separated. To finish the proof it suffices to show that f is universally closed. Let $V \to Y$ be an étale morphism where V is an affine scheme. It suffices to show that the base change $V \times_Y X \to V$ is universally closed, see Morphisms of Spaces, Lemma 03IT. Let

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & V \times_Y X & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & V & \longrightarrow & Y \end{array}$$

of algebraic spaces over S be a commutative diagram where A is a discrete valuation ring with fraction field K and where $\mathrm{Spec}(K) \to V \times_Y X$ maps to a point of codimension 0 of the algebraic space $V \times_Y X$. Since $V \times_Y X \to X$ is étale it follows that the image of $\mathrm{Spec}(K) \to X$ is a point of codimension 0 of X. Thus by (5) we obtain the longer of the two dotted arrows fitting into the diagram. Then of course we obtain the shorter one as well. It follows that our assumptions hold for the morphism $V \times_Y X \to V$ and we reduce to the case discussed in the next paragraph.

Aassume Y is a Noetherian affine scheme. In this case X is a separated Noetherian algebraic space (we already know f is separated) of finite type over Y. (In particular, the algebraic space X has a dense open subspace which is a scheme by Properties of Spaces, Proposition 06NH although strictly speaking we will not need this.) Choose a quasi-projective scheme X' over Y and a proper surjective morphism $X' \to X$ as in the weak form of Chow's lemma (Cohomology of Spaces, Lemma 089J). We may replace X' by the disjoint union of the irreducible components which dominate an irreducible component of X; details omitted. In particular, we may assume that generic points of the scheme X' map to points of codimension 0 of X (in this case these are exactly the generic points of X). We claim that $X' \to Y$ is proper. The claim implies X is proper over Y by Morphisms of Spaces, Lemma 08AJ. To prove this, according to Limits, Lemma 0H1X it suffices to prove that in every solid commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & X' & \longrightarrow & X \\ \downarrow & \overset{a}{\nearrow} & & \overset{b}{\nearrow} & \downarrow \\ \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

[1]There is a sharper formulation where in the existence part one only requires the dotted arrow exists after an extension of discrete valuation rings.

where A is a dvr with fraction field K and where K is the residue field of a generic point of X' we can find the dotted arrow a (we already know uniqueness as X' is separated). By assumption (5) we can find the dotted arrow b. Then the morphism $X' \times_{X,b} \mathrm{Spec}(A) \to \mathrm{Spec}(A)$ is a proper morphism of schemes and by the valuative criterion for morphisms of schemes we can lift b to the desired morphism a. $\square$